HostelFix ER Diagram

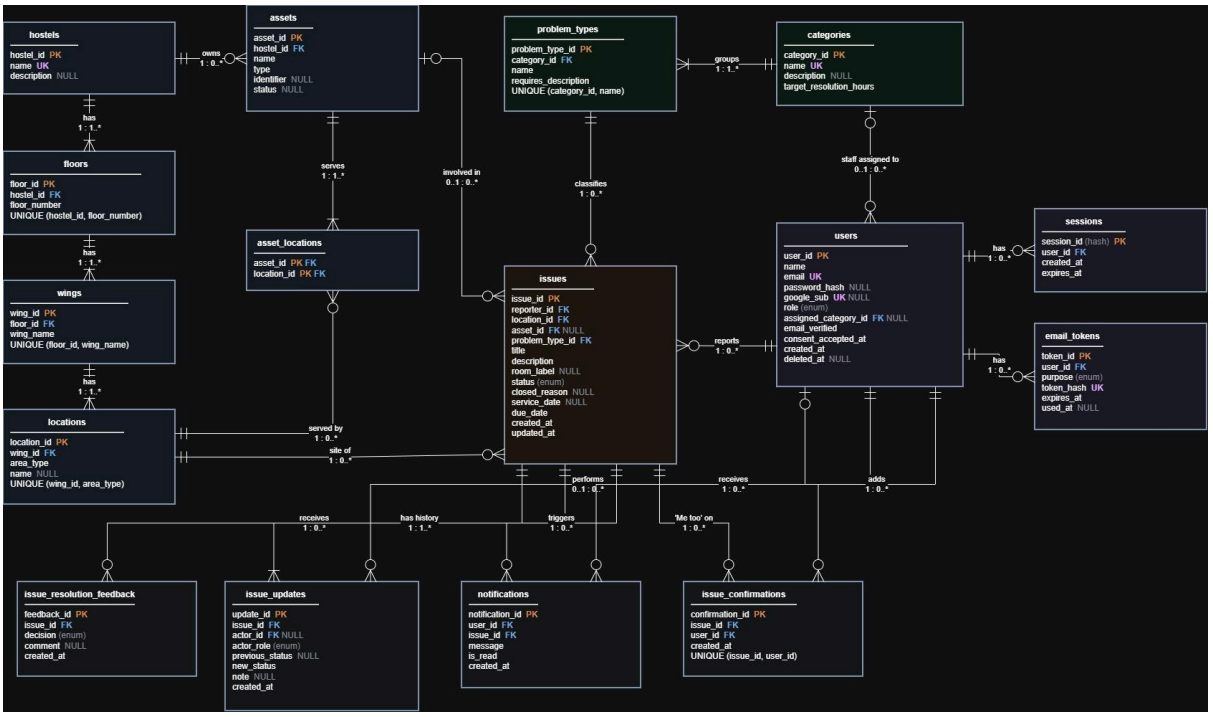


Fig 1 - ER Diagram

3NF notes

• issues has no category_id: the category is reached through problem_types.category_id.

• Feedback has no reporter_id: the reporter is issues.reporter_id.

• Hostel → floor → wing → location facts are stored once, one level per table.

• Account deletion anonymises the users row (deleted_at set), so issues.reporter_id stays valid.

Legend

PK : primary key

FK : foreign key

UK : unique

NULL : column is nullable; all others NOT NULL

1 : exactly one

0..1 : zero or one

0..* : zero or many

1..* : one or many

Cardinality on each line is read parent : child.

assets ↔ locations is many-to-many via asset_locations.

Index note

Each index below is tied to a page or query:

- **users(email) UNIQUE** — login lookup; blocks duplicate registration
- **users(google_sub) UNIQUE** — Google sign-in lookup
- **problem_types(category_id, name) UNIQUE** — cascading problem-type dropdown; one "Other" per category
- **hostels(name), floors(hostel_id, floor_number), wings(floor_id, wing_name), locations(wing_id, area_type) UNIQUE** — cascading hostel/floor/wing/area selects; no duplicate layout rows
- **asset_locations PK(asset_id, location_id) + INDEX(location_id)** — "which lifts serve this lobby" in Report an Issue
- **issues(reporter_id, created_at)** — My Reports and student Dashboard
- **issues(status, created_at)** — Community Issues default list and Staff Dashboard queue
- **issues(location_id, problem_type_id, status)** — similar-open-complaints check; location filters
- **issues(problem_type_id, status)** — Staff Dashboard per category; admin analytics by category/problem type
- **issues(due_date, status)** — overdue list on Staff and Admin Dashboards
- **issue_updates(issue_id, created_at)** — status timeline on Issue Details
- **issue_confirmations UNIQUE(issue_id, user_id)** — one "Me too" per user; its leading column also serves the per-issue count
- **issue_resolution_feedback(issue_id)** — feedback and reopen requests on Issue Details
- **notifications(user_id, is_read, created_at)** — notification bell and Dashboard, unread first
- **email_tokens(token_hash) UNIQUE; email_tokens(user_id, purpose)** — verify / reset link lookup
- **sessions(user_id); sessions(expires_at)** — end all sessions on password reset; clean up expired sessions